

subjects (Baenninger & Greco, 1991). The questionnaire results are shown in Table 1. Particular situations reported by some as conducive to yawning included lectures (750/o), being tired or lacking sleep (32%), being under stress, or seeing other people yawn. To us it is noteworthy that there is no real consensus about situations or stimuli associated with yawning.

Table 1 Yawning Questionnaire

Situation	Likely	Unlikely	Unknown
Listening to speech or lecture	118	27	12
Sitting in a traffic jam	60	82	15
Waiting for a train or bus	66	60	26
Driving at night on lonely highway	90	47	15
Driving on sunny day (no traffic)	19	113	20
While giving a speech or lecture	4	140	8
Waiting to begin a competitive event	18	120	14
Being interviewed for a job	4	136	12
Lying in bed before falling asleep	83	57	12
Getting out of bed in the morning	121	29	2

Yawn more in: morning 38(25%) afternoon 64(42%) evening 40(26%) noresponse 10

Yawn more while:

tired	25
lecture/class	29
lack of sleep	15
stress	9
after a meal	8
see others yawn	8
working	7
study at night	4
in church	4

GENERAL DISCUSSION

Although there were areas of agreement between the two studies some discrepancies emerged that may reflect differences between keeping a log and responding to a survey: The former represents what subjects actually do, and the latter represents what they believe they do. For example, the log data indicated no relationship between sleep duration and yawning frequency during the following day; survey results indicated that respondents believe that there is a relationship.

Survey respondents believed that they yawn more in the afternoon than in either the morning or evening, but the log results flatly contradict this belief because afternoon hours were when the percentages of daily yawns fell to their lowest levels. Survey respondents believed that lying in bed at night and arising in the morning were both situations where yawning was frequent. Partly supporting this finding, the logs in Study 1 showed that the first hour after waking and the last hour before bed were the times when most yawns occurred, although the logs indicated that it is the activities that accompany retiring to bed and arising in the morning that are associated with yawning. In any event, this temporal variation supports findings of Provine, Hamernik, and Curchak (1987) in humans and of Anias, Holmgren, Urba-Holmgren, and Eguibar (1984) in rats. In rats of a strain selectively bred for a high frequency of yawning (Urba-